Building a Question Answering System from Scratch Using the Stanford SQuAD Dataset

1 Problem Statement

Question Answering (QA) is a Natural Language Processing task where a system reads a paragraph and returns the exact answer to a given question.

Example: Paragraph: "Hyderabad was founded in 1591 by Muhammad Quli Qutb Shah. It is the capital of Telangana."

Question: "When was Hyderabad founded?"

Answer: 1591

This is called Machine Reading Comprehension. It is harder than it looks | the question and the paragraph rarely use the same words, so the model must understand the meaning behind both and connect them.

In this project we build a transformer-based QA system using PyTorch. The model takes a question and a paragraph as input and identi es the exact span of text that answers the question.

We train the system in three stages. First, the model is pretrained on Wikipedia text to learn general English language patterns. Second, it is ne-tuned on the SQuAD 1.1 dataset, which contains 87,599 question-answer pairs from Wikipedia, to learn extractive question answering. As an extension, a decoder is added so the model can generate a complete sentence answer instead of extracting a word directly from the paragraph. The system is evaluated using Exact Match and F1 score on the SQuAD validation set.

2 System Architecture

The architecture of the proposed system consists of the following components:

Input Layer

The input consists of a question and a context paragraph combined into a single token sequence.

Embedding Layer

This layer converts tokens into dense vector representations using token embeddings, positional embeddings, and segment embeddings.

Transformer Encoder

The core of the model consists of stacked transformer encoder layers. Each layer includes:

- Multi-head self-attention mechanism
- Feedforward neural network
- Layer normalization
- Residual connections